

CSE411 - Artificial Intelligence | Final Examination, Spring 2025

Questions 2 & 3 — Full Solutions

QUESTION 2

Question 2(a) — [3 Marks | CO2]

Discuss general logic characteristics with the help of entailment.

Answer:

Logic is a formal system used in AI for knowledge representation and reasoning. The general characteristics of logic are discussed below:

1. Syntax

Syntax defines the structure or grammar of valid sentences in a logic system. It specifies how symbols can be combined to form well-formed formulas (WFFs). For example, in propositional logic, $p \wedge q$ is syntactically valid, while $\wedge p q \wedge$ is not.

2. Semantics

Semantics defines the meaning of sentences — i.e., how truth values are assigned to sentences based on interpretations. A sentence is **true** or **false** under a given model/world.

3. Entailment ($\models$)

Entailment is the most important logical characteristic. A knowledge base **KB** entails a sentence α (written $KB \models \alpha$) if and only if **in every model where KB is true, α is also true.**

$KB \models \alpha$ means: There is no possible world where KB is true but α is false.

Example:

- $KB = \{p \rightarrow q, p\}$
- $\alpha = q$

- In every model where $p \rightarrow q$ and p are true, q must be true.
- Therefore, $KB \models q$ ✓

4. Soundness

An inference procedure is **sound** if it derives only sentences that are actually entailed. It never produces false conclusions.

5. Completeness

An inference procedure is **complete** if it can derive every sentence that is entailed by the KB. It misses nothing.

6. Validity

A sentence is **valid** (a tautology) if it is true in all possible models/worlds. Example: $p \vee \neg p$ is always true.

7. Satisfiability

A sentence is **satisfiable** if there exists at least one model in which it is true.

Summary Table:

Characteristic	Description
Syntax	Grammar rules of the language
Semantics	Meaning / truth assignment
Entailment	$KB \models \alpha$ if α is true in all models of KB
Soundness	Only derives true conclusions
Completeness	Derives all true conclusions
Validity	True in ALL models
Satisfiability	True in SOME model

Question 2(b) — [2 Marks | CO2]

Identify KB entails sentence ' α ' if and only if ' α ' is true in all worlds where KB is true.
Given:

- $KB = (p \wedge q), (\neg p \wedge \neg r)$, and $\alpha = \neg p \wedge (q \vee \neg r)$
- Apply on the truth table.

Answer:

We need to check whether $\mathbf{KB} \models \alpha$, i.e., in every row where all KB sentences are true, is α also true?

KB sentences:

- $\mathbf{KB}_1 = p \wedge q$
- $\mathbf{KB}_2 = \neg p \wedge \neg r$
- $\alpha = \neg p \wedge (q \vee \neg r)$

Full Truth Table:

p	q	r	$p \wedge q$	$\neg p$	$\neg r$	$\neg p \wedge \neg r$	$q \vee \neg r$	$\neg p \wedge (q \vee \neg r)$	KB True?	α True?
T	T	T	T	F	F	F	T	F	F	F
T	T	F	T	F	T	F	T	F	F	F
T	F	T	F	F	F	F	F	F	F	F
T	F	F	F	F	T	F	F	F	F	F
F	T	T	F	T	F	F	T	T	F	T
F	T	F	F	T	T	T	T	T	F	T
F	F	T	F	T	F	F	F	F	F	F
F	F	F	F	T	T	T	F	F	F	F

Analysis:

For KB to be true, **both** $\mathbf{KB}_1$ ($p \wedge q$) and $\mathbf{KB}_2$ ($\neg p \wedge \neg r$) must be true simultaneously.

- $\mathbf{KB}_1 = p \wedge q$ requires $\mathbf{p = T}$
- $\mathbf{KB}_2 = \neg p \wedge \neg r$ requires $\mathbf{p = F}$

These two conditions **contradict each other** — p cannot be both T and F at the same time.

➡ Therefore, **there is no model (row) where KB is true.**

Conclusion:

Since KB is **unsatisfiable** (false in all worlds), the condition " α is true in every world where KB is true" is **vacuously satisfied**.

∴ $\mathbf{KB} \models \alpha$ ✓ (Entailment holds vacuously)

Question 2(c) — [5 Marks | CO2]

Predict truth table on the statement considering propositional symbols: "I make you Appetizer, Dinner, and Dessert"

Answer:

Define Propositional Symbols:

- **A** = I make you Appetizer
- **D** = I make you Dinner
- **De** = I make you Dessert

The full statement is: **$A \wedge D \wedge De$** (I make you Appetizer AND Dinner AND Dessert)

Truth Table for $A \wedge D \wedge De$:

A	D	De	$A \wedge D$	$A \wedge D \wedge De$
T	T	T	T	T
T	T	F	T	F
T	F	T	F	F
T	F	F	F	F
F	T	T	F	F
F	T	F	F	F
F	F	T	F	F
F	F	F	F	F

Interpretation:

- The statement "**I make you Appetizer, Dinner, and Dessert**" is only **TRUE** when ALL THREE dishes are made (row 1).
- In all other 7 combinations, the statement is **FALSE**.
- This demonstrates the **conjunctive** (AND) nature — every component must be satisfied.

Additional Logical Properties:

- The statement is **satisfiable** (true in at least one model — row 1)
 - The statement is **not valid** (not true in all models)
 - **Negation:** $\neg(A \wedge D \wedge De) = \neg A \vee \neg D \vee \neg De$ (by De Morgan's Law) $\rightarrow$ "I did NOT make at least one of Appetizer, Dinner, or Dessert"
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QUESTION 3

Question 3(a) — [5 Marks | CO3]

Demonstrate uncertainty reasons which create ambiguity in an Autonomous Agent.

Answer:

An **Autonomous Agent** is a system that perceives its environment and takes actions to achieve goals. However, in real-world environments, agents face **uncertainty** that creates ambiguity in decision-making.

Definition of Uncertainty:

Uncertainty arises when an agent **lacks complete, correct, or sufficient information** about the world, the effects of its actions, or other agents' behaviors.

Reasons for Uncertainty in Autonomous Agents:

1. 🔍 Partial Observability

The agent cannot observe the complete state of the environment at all times.

Example: A self-driving car cannot see around corners or through other vehicles. It doesn't know if a pedestrian is hidden behind a parked car.

➡ **Ambiguity:** The agent is unsure of the current world state.

2. ⚙️ Non-deterministic Actions (Noisy Actuators)

The outcome of an action may not always be what was intended due to physical limitations or mechanical errors.

Example: A robot arm commanded to pick up a cup may slip or grasp it improperly.

→ **Ambiguity:** The agent is unsure what will happen after taking an action.

3. Noisy / Unreliable Sensors

Sensor readings may be inaccurate, delayed, or corrupted by environmental noise.

Example: A medical diagnosis robot may receive incorrect readings from a faulty thermometer or ECG sensor.

→ **Ambiguity:** The agent is unsure whether its perception reflects reality.

4. Dynamic and Changing Environment

The environment changes over time, sometimes while the agent is still reasoning or executing a plan.

Example: A drone navigating outdoors faces changing wind speed, rain, or moving obstacles.

→ **Ambiguity:** A plan that was optimal a moment ago may now be wrong.

5. Uncertainty About Other Agents

In multi-agent environments, an agent cannot perfectly predict the behavior of other agents (human or AI).

Example: In a game or traffic system, another driver's next move is unpredictable.

→ **Ambiguity:** The agent is unsure how others will behave.

6. Incomplete or Incorrect Knowledge Base

The agent's prior knowledge may be outdated, incomplete, or incorrect.

Example: A medical AI trained on old data may give wrong diagnoses for new diseases.

→ **Ambiguity:** The agent reasons from incorrect premises.

7. 🕒 Temporal Uncertainty

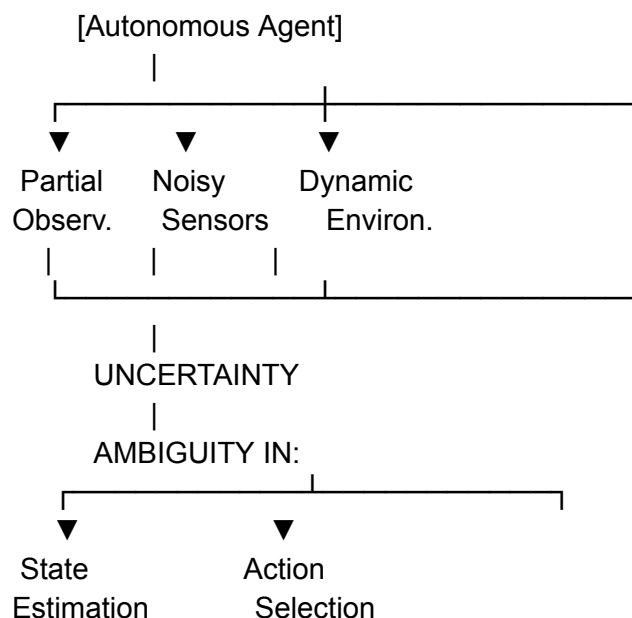
Events happen at uncertain times; the agent may not know when something occurred or will occur.

Example: An AI scheduler doesn't know exactly when a user will be free.

How Autonomous Agents Handle Uncertainty:

Technique	Description
Probability Theory	Assigns probabilities to possible states
Bayesian Networks	Models conditional dependencies
Fuzzy Logic	Handles partial truth values
Markov Decision Processes (MDP)	Handles sequential decisions under uncertainty
Belief States	Agent maintains a set of possible current states

Summary Diagram:



Question 3(b) — [5 Marks | CO3]

The full joint distribution on the following table describes the probabilities of combinations of vehicle types and vehicle owners. Applying these probabilities, analyze the following questions.

Given Table:

Age of owner	Car	Minivan	SUV
Under 30	0.15	0.05	0.10
Between 30 and 50	0.10	0.15	0.10
Over 50	0.15	0.15	0.05

Verification (Sum of all probabilities):

$$0.15 + 0.05 + 0.10 + 0.10 + 0.15 + 0.10 + 0.15 + 0.15 + 0.05 = 1.00 \checkmark$$

i) Determine the probability that a vehicle owner is under 30 years old.

P(age < 30) = Sum of all entries in the "Under 30" row

$$P(\text{age} < 30) = P(\text{Car} \wedge \text{age} < 30) + P(\text{Minivan} \wedge \text{age} < 30) + P(\text{SUV} \wedge \text{age} < 30)$$

$$P(\text{age} < 30) = 0.15 + 0.05 + 0.10$$

$$\therefore P(\text{age} < 30) = 0.30$$

ii) Determine P(vehicle type = Minivan | age of owner is under 30)

Using **Bayes' / Conditional Probability** formula:

$$P(\text{Minivan} \mid \text{age} < 30) = \frac{P(\text{Minivan} \cap \text{age} < 30)}{P(\text{age} < 30)}$$

From the table:

- $P(\text{Minivan} \cap \text{age} < 30) = 0.05$
- $P(\text{age} < 30) = 0.30$ (calculated above)

$$P(\text{Minivan} \mid \text{age} < 30) = \frac{0.05}{0.30} = \frac{5}{30} = \frac{1}{6}$$

$$\therefore P(\text{Minivan} \mid \text{age} < 30) \approx 0.1667 \text{ (16.67\%)}$$

iii) Determine $P(\text{vehicle type} = \text{SUV} \vee \text{age of owner is over } 50)$

Using the **Addition Rule (OR rule)**:

$$P(\text{SUV} \cup \text{age} > 50) = P(\text{SUV}) + P(\text{age} > 50) - P(\text{SUV} \cap \text{age} > 50)$$

Step 1: Find $P(\text{SUV})$

$$P(\text{SUV}) = P(\text{SUV} \wedge \text{age} < 30) + P(\text{SUV} \wedge 30 \leq \text{age} \leq 50) + P(\text{SUV} \wedge \text{age} > 50)$$

$$P(\text{SUV}) = 0.10 + 0.10 + 0.05 = \mathbf{0.25}$$

Step 2: Find $P(\text{age} > 50)$

$$P(\text{age} > 50) = P(\text{Car} \wedge \text{age} > 50) + P(\text{Minivan} \wedge \text{age} > 50) + P(\text{SUV} \wedge \text{age} > 50)$$

$$P(\text{age} > 50) = 0.15 + 0.15 + 0.05 = \mathbf{0.35}$$

Step 3: Find $P(\text{SUV} \cap \text{age} > 50)$

$$P(\text{SUV} \cap \text{age} > 50) = \mathbf{0.05} \text{ (directly from table)}$$

Step 4: Apply Addition Rule:

$$P(\text{SUV} \cup \text{age} > 50) = 0.25 + 0.35 - 0.05$$

$$\therefore P(\text{SUV} \vee \text{age} > 50) = \mathbf{0.55 \text{ (55\%)}}$$



End of Question 2 and Question 3